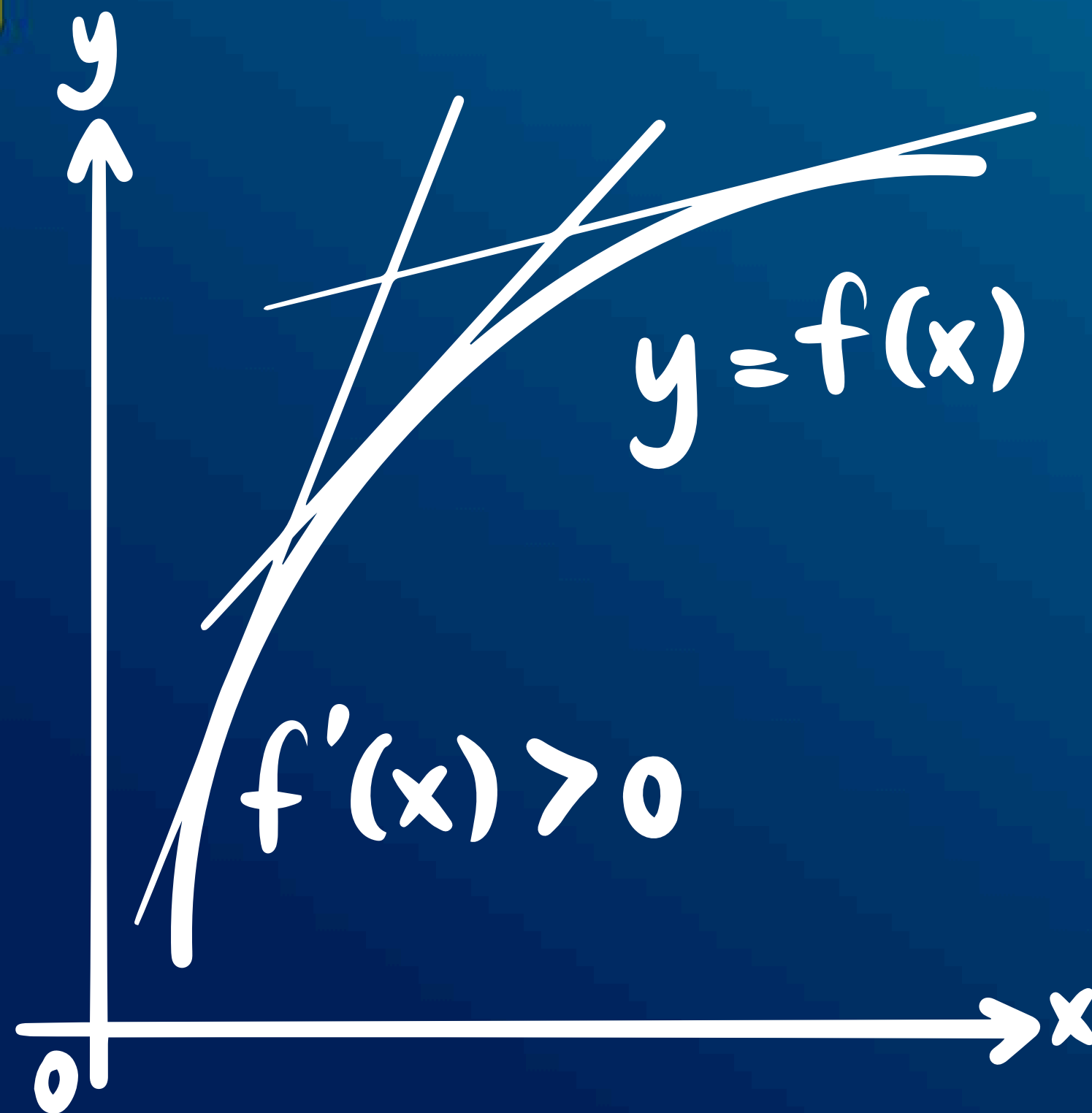
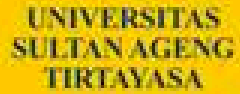




Calculus 1

By : Ferdian Bangkit Wijaya, S.Stat, M.Si



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Jujur Adil Wibawa Amanah Religius Akuntabel

(Definition, Limit : Algebra, Trigonometric, Infinite)

The image is a complex collage of mathematical content. At the center, a white rectangular box with a black border contains the limit definition of a function: $\lim_{x \rightarrow a} f(x) = f(a)$. A black arrow points from the top of this box towards the top-left corner of the image. The background is filled with various mathematical elements:

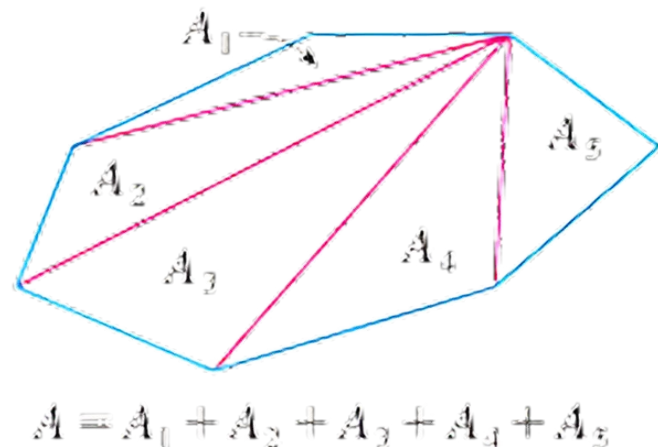
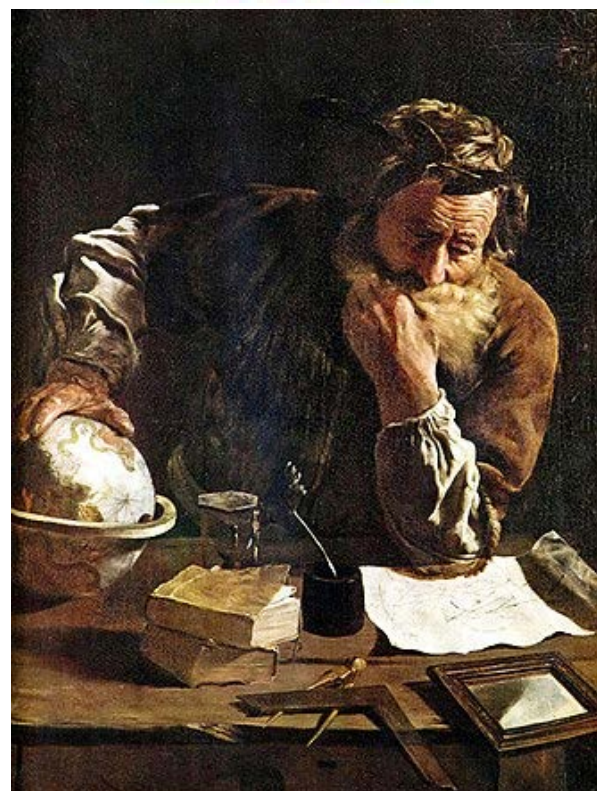
- Formulas:**
 - Trigonometric: $\cos 2A = \cos^2 A - \sin^2 A$, $\sin 2A = 2 \sin A \cos A$, $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$, $\cos^2 A = \frac{1 + \cos 2A}{2}$, $\sin^2 A = \frac{1 - \cos 2A}{2}$.
 - Calculus: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$, $\frac{d}{dx} \ln x = \frac{1}{x}$, $\frac{d}{dx} e^x = e^x$, $\frac{d}{dx} \sin x = \cos x$, $\frac{d}{dx} \cos x = -\sin x$, $\frac{d}{dx} \tan x = \sec^2 x$, $\frac{d}{dx} \cot x = -\csc^2 x$, $\frac{d}{dx} \sec x = \sec x \tan x$, $\frac{d}{dx} \csc x = -\csc x \cot x$.
 - Algebra: $y = \frac{\Delta x}{\Delta z}$, $y = \frac{\Delta x}{\Delta z} + 2$, $y = \frac{\Delta x}{\Delta z} - 1$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{2}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{2}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{3}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{3}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{4}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{4}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{5}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{5}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{6}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{6}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{7}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{7}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{8}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{8}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{9}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{9}$, $y = \frac{\Delta x}{\Delta z} + \frac{1}{10}$, $y = \frac{\Delta x}{\Delta z} - \frac{1}{10}$.
 - Geometry: $\sin^2 \theta + \cos^2 \theta = 1$, $\tan^2 \theta + 1 = \sec^2 \theta$, $\cot^2 \theta + 1 = \csc^2 \theta$, $\sec^2 \theta = 1 + \tan^2 \theta$, $\csc^2 \theta = 1 + \cot^2 \theta$.
 - Series: $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$, $\sum_{n=0}^{\infty} x^{2n} = \frac{1}{1-x^2}$, $\sum_{n=0}^{\infty} x^{2n+1} = \frac{x}{1-x^2}$, $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$, $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \cosh x$, $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = \sinh x$.
- Graphs:**
 - Graphs of $y = \sin x$, $y = \cos x$, $y = \tan x$, $y = \cot x$, $y = \sec x$, $y = \csc x$.
 - Graphs of $y = e^x$, $y = \ln x$.
 - Graphs of $y = x^2$, $y = x^3$, $y = x^4$, $y = x^5$, $y = x^6$, $y = x^7$, $y = x^8$, $y = x^9$, $y = x^{10}$.
 - Graphs of $y = \frac{1}{x}$, $y = \frac{1}{x^2}$, $y = \frac{1}{x^3}$, $y = \frac{1}{x^4}$, $y = \frac{1}{x^5}$, $y = \frac{1}{x^6}$, $y = \frac{1}{x^7}$, $y = \frac{1}{x^8}$, $y = \frac{1}{x^9}$, $y = \frac{1}{x^{10}}$.
 - Graphs of $y = \sqrt{x}$, $y = \sqrt[3]{x}$, $y = \sqrt[4]{x}$, $y = \sqrt[5]{x}$, $y = \sqrt[6]{x}$, $y = \sqrt[7]{x}$, $y = \sqrt[8]{x}$, $y = \sqrt[9]{x}$, $y = \sqrt[10]{x}$.



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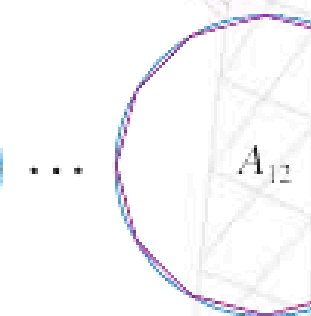
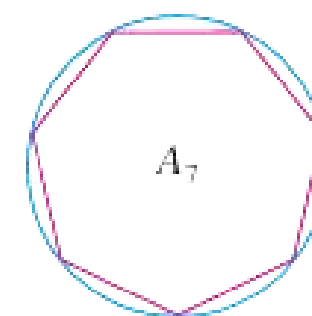
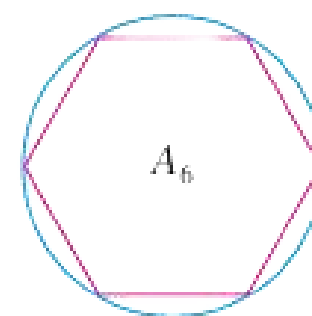
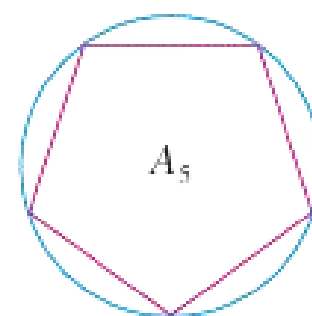
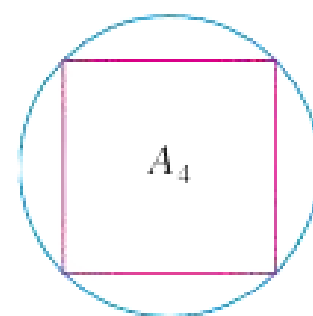
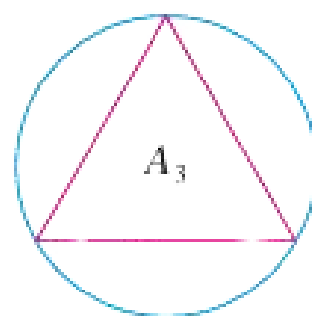
Limit

Historical



$$A = A_1 + A_2 + A_3 + A_4 + A_5$$

Ancient Greek Era
2500 years ago
By : Archimedes
“Method of Exhaustions”



$$A = \lim_{n \rightarrow \infty} A_n$$



Revolution Era
17th Century
By : Isaac Newton
“Differential & Integral for movement
of planets and physical objects”

Revolution Era
17th Century
By : Gottfried Leibniz
“Differential & Integral Notation”



Limit

Introduction to Limits

Definition Intuitive Meaning of Limit

To say that $\lim_{x \rightarrow c} f(x) = L$ means that when x is near but different from c then $f(x)$ is near L .

Graph Solutions

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3$$

Algebra Solutions

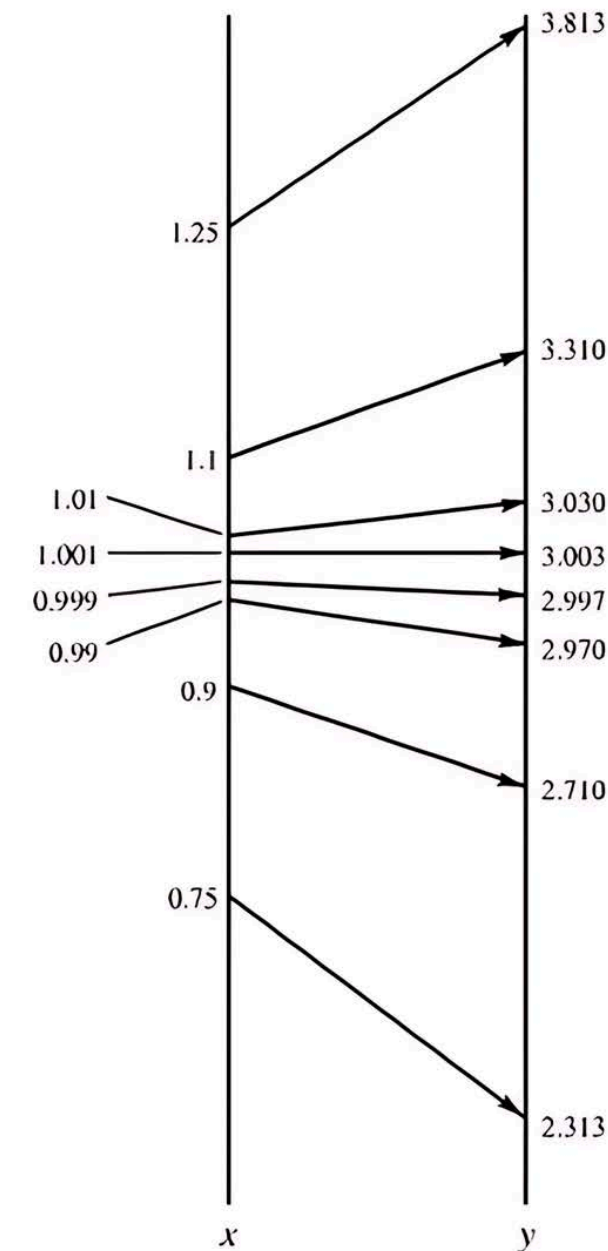
$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x - 1)(x^2 + x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x^2 + x + 1) = 1^2 + 1 + 1 = 3 \end{aligned}$$

Note that $(x - 1)/(x - 1) = 1$ as long as $x \neq 1$

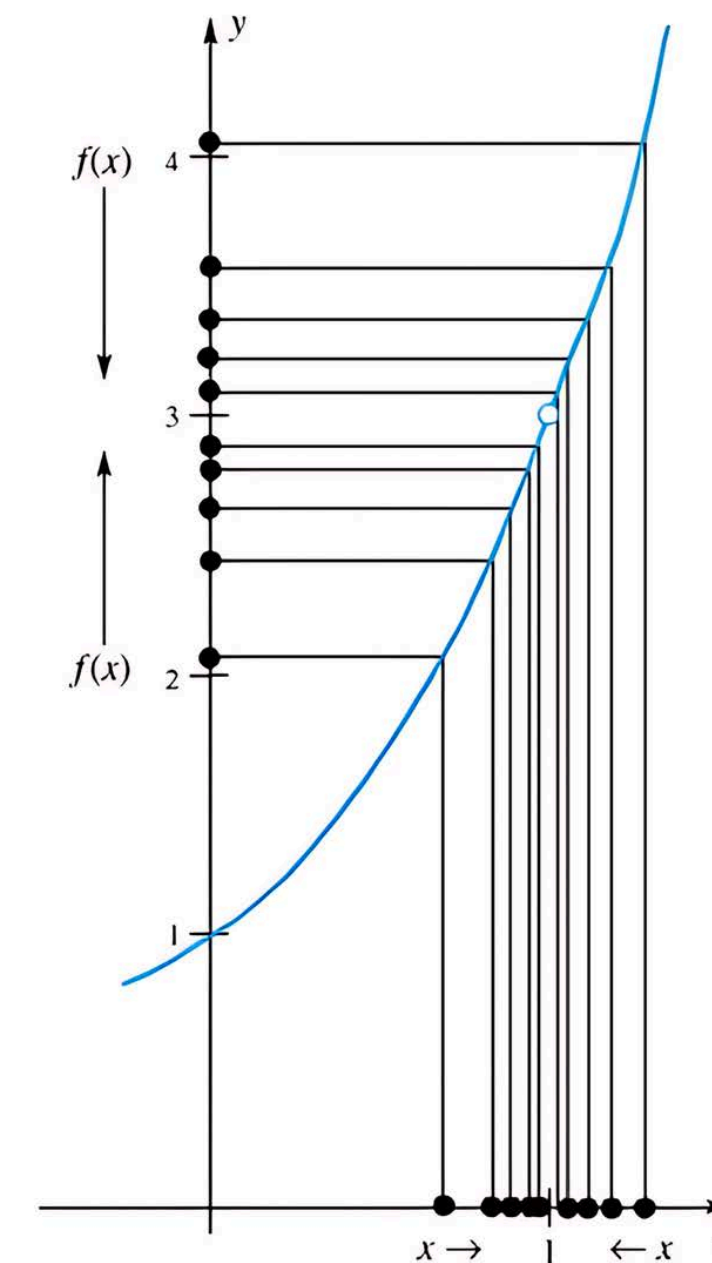
EXAMPLE 1

x	$y = \frac{x^3 - 1}{x - 1}$
1.25	3.813
1.1	3.310
1.01	3.030
1.001	3.003
↓	↓
1.000	?
↑	↑
0.999	2.997
0.99	2.970
0.9	2.710
0.75	2.313

Table
of values



Schematic
diagram



Graph of $y = f(x) = \frac{x^3 - 1}{x - 1}$

Graph Solutions

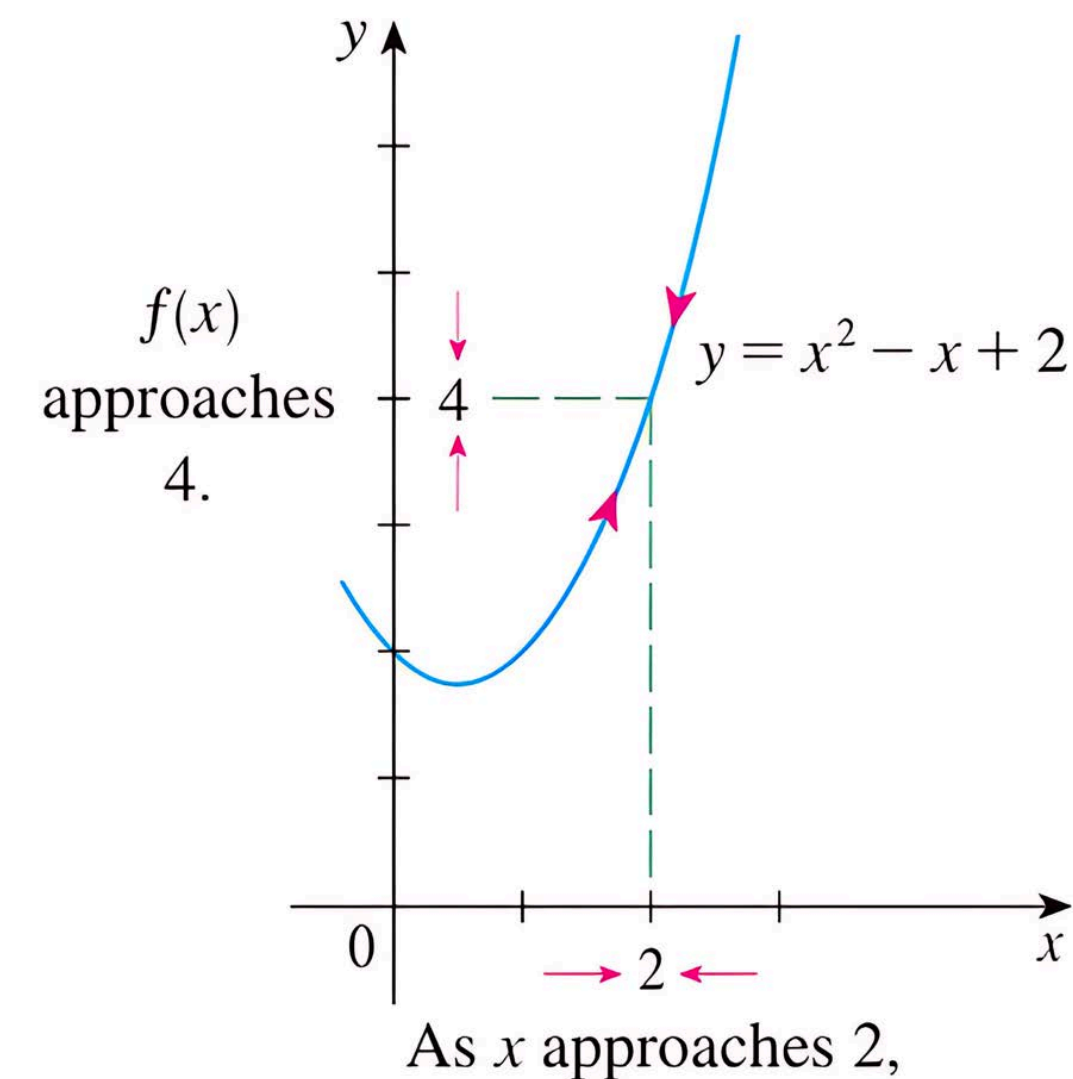
$$\lim_{x \rightarrow 2} (x^2 - x + 2) = 4$$

Algebra Solutions

$$\lim_{x \rightarrow 2} (x^2 - x + 2) = 4$$

EXAMPLE 2

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001



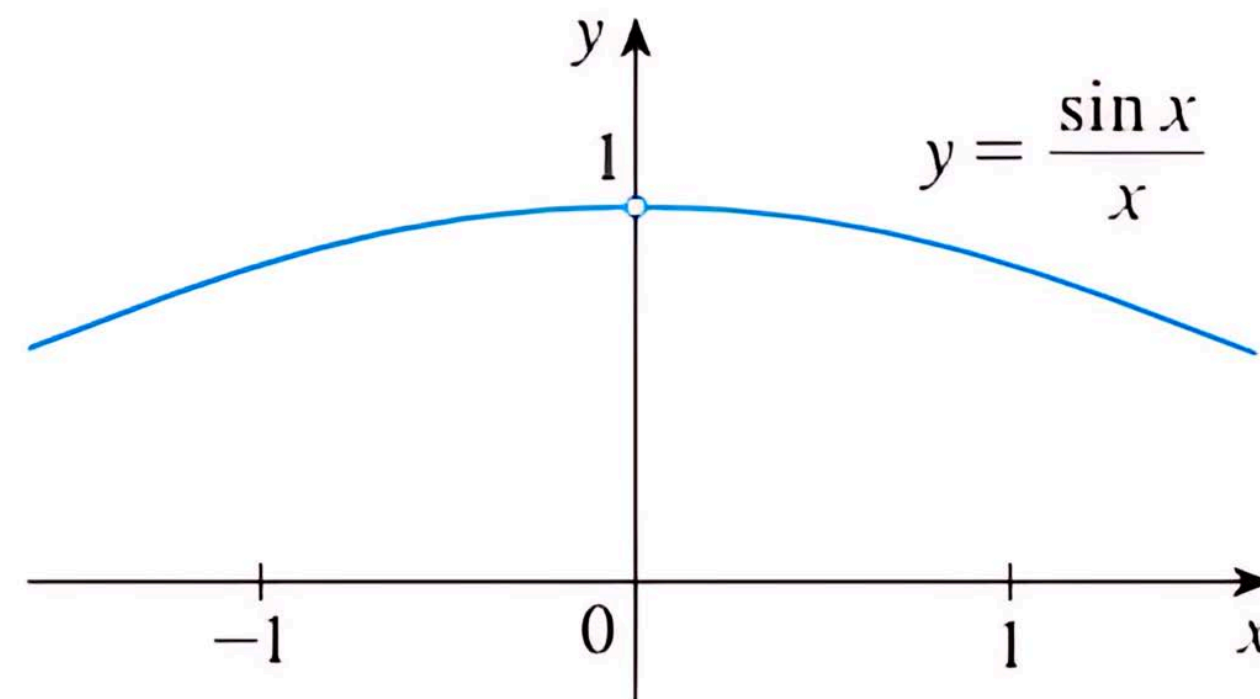
EXAMPLE 3

Question

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

x in RAD measure

Answered



x	$\frac{\sin x}{x}$
± 1.0	0.84147098
± 0.5	0.95885108
± 0.4	0.97354586
± 0.3	0.98506736
± 0.2	0.99334665
± 0.1	0.99833417
± 0.05	0.99958339
± 0.01	0.99998333
± 0.005	0.99999583
± 0.001	0.99999983



Limit

Introduction to Limits



EXAMPLE 4

Question

$$\lim_{t \rightarrow -1} (t^2 - 1)$$

Answered

$$\lim_{t \rightarrow -1} (t^2 - 1) = ((-1)^2 - 1) = 0$$

EXAMPLE 5

Question

$$\lim_{x \rightarrow -2} (x^2 + 2t - 1)$$

Answered

$$\lim_{x \rightarrow -2} (x^2 + 2t - 1) = (-2)^2 + 2t - 1 = 3 + 2t$$

EXAMPLE 6

Question

$$\lim_{t \rightarrow -1} (t^2 - x^2)$$

Answered

$$\lim_{t \rightarrow -1} (t^2 - x^2) = ((-1)^2 - x^2) = 1 - x^2$$

EXAMPLE 7

Question

$$\lim_{x \rightarrow -1} \frac{x^3 - 4x^2 + x + 6}{x + 1}$$

Answered

$$= \lim_{x \rightarrow -1} \frac{(x+1)(x^2 - 5x + 6)}{x+1} = (-1)^2 - 5(-1) + 6 = 12$$

EXAMPLE 8

Question

$$\lim_{x \rightarrow 0} \frac{x^4 + 2x^3 - x^2}{x^2}$$

Answered

$$= \lim_{x \rightarrow 0} (x^2 + 2x - 1) = -1$$

EXAMPLE 9

Question

$$\lim_{x \rightarrow -t} \frac{x^2 - t^2}{x + t}$$

Answered

$$\lim_{x \rightarrow -t} \frac{(x+t)(x-t)}{x+t} = -t - t = -2t$$

EXAMPLE 10

Answered

Question

$$\lim_{x \rightarrow 0} \frac{\sin x}{2x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{2x} = 0.5$$

x	$\frac{\sin x}{2x}$
1.	0.420735
0.1	0.499167
0.01	0.499992
0.001	0.49999992
-1.	0.420735
-0.1	0.499167
-0.01	0.499992
-0.001	0.49999992

EXAMPLE 11

Answered

Question

$$\lim_{x \rightarrow 3} \frac{x - \sin(x-3) - 3}{x-3}$$

$$\lim_{x \rightarrow 3} \frac{x - \sin(x-3) - 3}{x-3} = 0$$

x	$\frac{x - \sin(x-3) - 3}{x-3}$
4.	0.158529
3.1	0.00166583
3.01	0.0000166666
3.001	1.66667×10^{-7}
2.	0.158529
2.9	0.00166583
2.99	0.0000166666
2.999	1.66667×10^{-7}

Limit

Introduction to Limits (Time to participate)

Sketch the graph of

$$f(x) = \begin{cases} x + 1 & \text{if } x \leq -1 \\ x^2 & \text{if } x > -1 \end{cases}$$

1

Sketch the graph of

$$f(x) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 1 & \text{if } |x| > 1 \end{cases}$$

2

find the indicated limit

$$\lim_{x \rightarrow 3} \frac{x^4 - 18x^2 + 81}{(x - 3)^2}$$

3

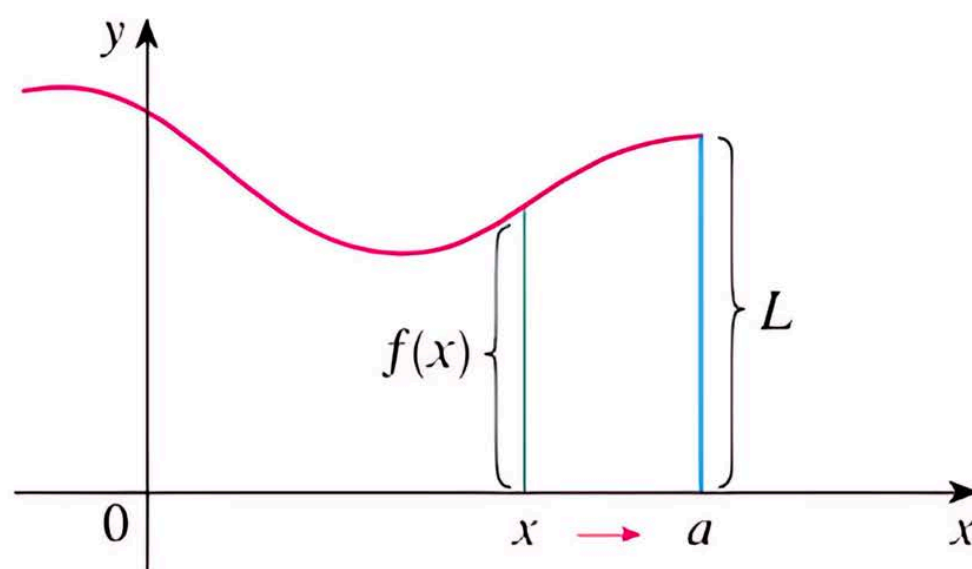
Limit

One Side Limits

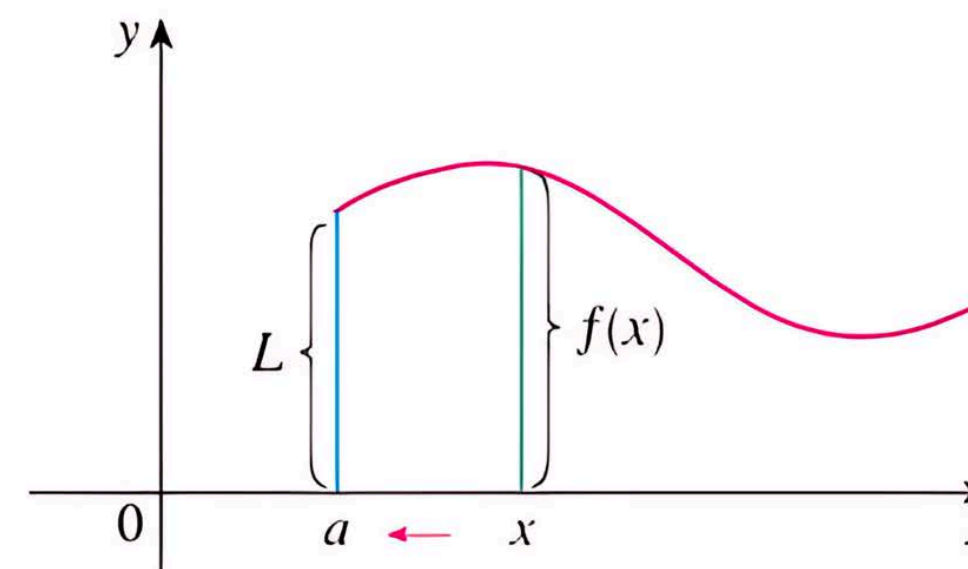
2 Definition of One-Sided Limits We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the **left-hand limit of $f(x)$ as x approaches a** [or the **limit of $f(x)$ as x approaches a from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x *less than* a .



(a) $\lim_{x \rightarrow a^-} f(x) = L$



(b) $\lim_{x \rightarrow a^+} f(x) = L$

Limit

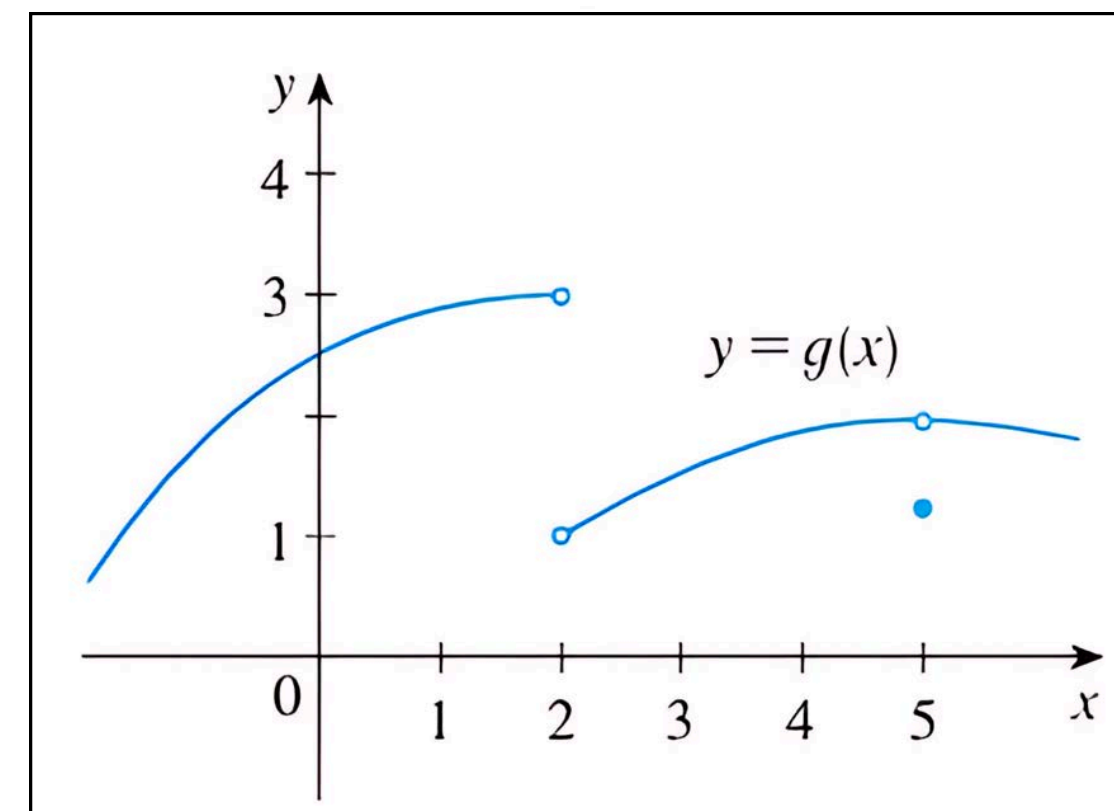
Definitions of Limits

1 Intuitive Definition of a Limit Suppose $f(x)$ is defined when x is near the number a . (This means that f is defined on some open interval that contains a , except possibly at a itself.) Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

and say “the limit of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a .



3 $\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$

(a) $\lim_{x \rightarrow 2^-} g(x) = 3$ (b) $\lim_{x \rightarrow 2^+} g(x) = 1$

(c) $\lim_{x \rightarrow 2} g(x)$ does not exist

(d) $\lim_{x \rightarrow 5^-} g(x) = 2$ (e) $\lim_{x \rightarrow 5^+} g(x) = 2$

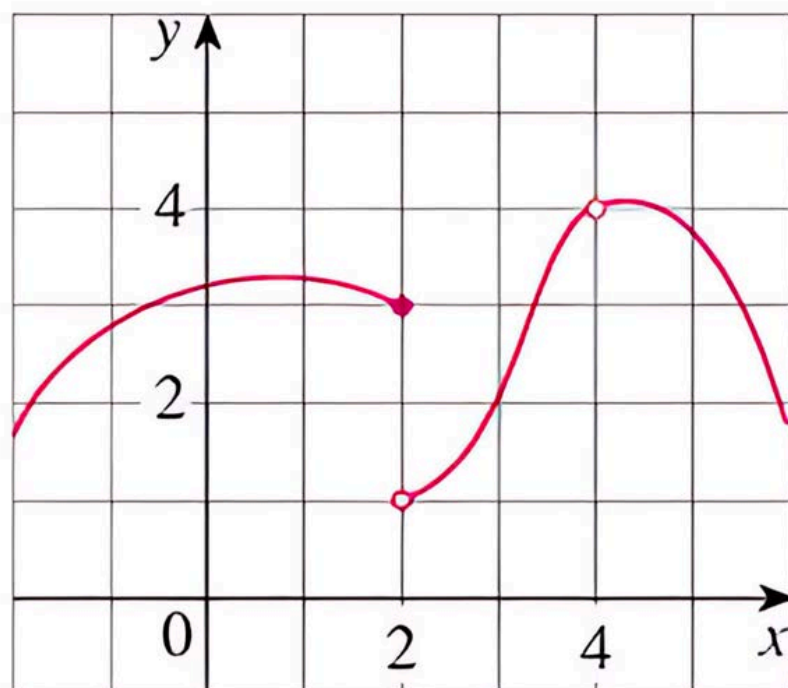
(f) $\lim_{x \rightarrow 5} g(x)$

$\lim_{x \rightarrow 5} g(x) = 2$ Despite this fact, notice that $g(5) \neq 2$.

Limit

Definitions of Limits

EXAMPLE 1



Question

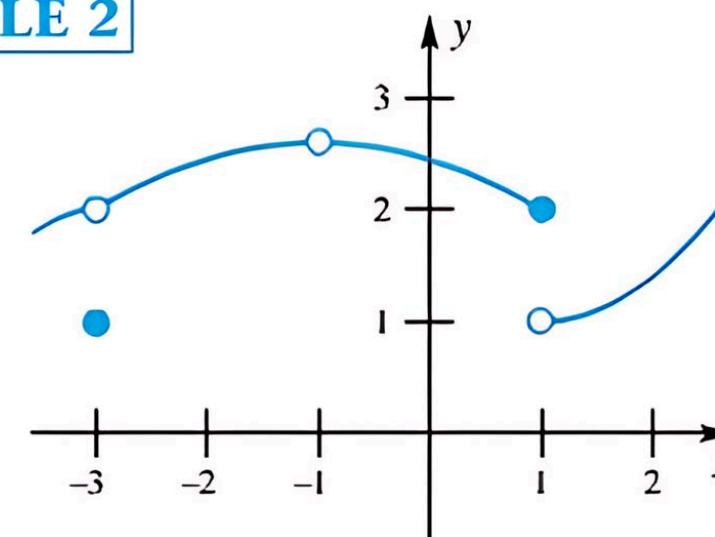
Use the given graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.

- | | | |
|-------------------------------------|-------------------------------------|-----------------------------------|
| (a) $\lim_{x \rightarrow 2^-} f(x)$ | (b) $\lim_{x \rightarrow 2^+} f(x)$ | (c) $\lim_{x \rightarrow 2} f(x)$ |
| (d) $f(2)$ | (e) $\lim_{x \rightarrow 4} f(x)$ | (f) $f(4)$ |

Answered

- (a) $\lim_{x \rightarrow 2^-} f(x) = 3.$
 (b) $\lim_{x \rightarrow 2^+} f(x) = 1.$
 (c) $\lim_{x \rightarrow 2} f(x)$ does not exist
 (d) $f(2) = 3.$
 (e) $\lim_{x \rightarrow 4} f(x) = 4.$
 (f) $f(4)$ does not exist.

EXAMPLE 2



Question

- | | | |
|-------------------------------------|-------------------------------------|--------------------------------------|
| (a) $\lim_{x \rightarrow -3} f(x)$ | (b) $f(-3)$ | (c) $f(-1)$ |
| (d) $\lim_{x \rightarrow -1} f(x)$ | (e) $f(1)$ | (f) $\lim_{x \rightarrow 1} f(x)$ |
| (g) $\lim_{x \rightarrow 1^-} f(x)$ | (h) $\lim_{x \rightarrow 1^+} f(x)$ | (i) $\lim_{x \rightarrow -1^+} f(x)$ |

Answered

- | | |
|---|---|
| a. $\lim_{x \rightarrow -3} f(x) = 2$ | f. $\lim_{x \rightarrow 1} f(x)$ does not exist. |
| b. $f(-3) = 1$ | g. $\lim_{x \rightarrow 1^-} f(x) = 2$ |
| c. $f(-1)$ does not exist. | h. $\lim_{x \rightarrow 1^+} f(x) = 1$ |
| d. $\lim_{x \rightarrow -1} f(x) = \frac{5}{2}$ | i. $\lim_{x \rightarrow -1^+} f(x) = \frac{5}{2}$ |
| e. $f(1) = 2$ | |

Limit

Limit of Algebra

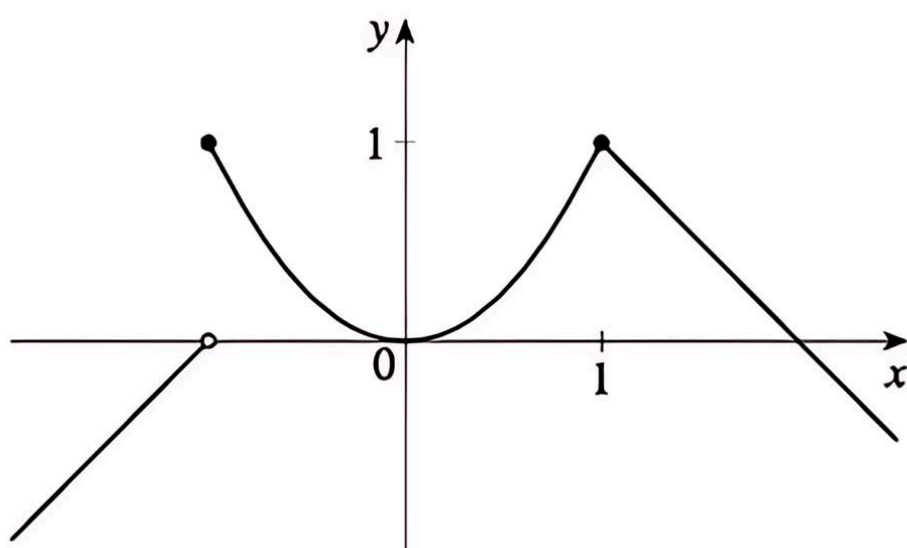
EXAMPLE 3

Question

Sketch the graph of the function and use it to determine the values of a for which $\lim_{x \rightarrow a} f(x)$ exists.

$$f(x) = \begin{cases} 1 + x & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ 2 - x & \text{if } x \geq 1 \end{cases}$$

Answered



we see that $\lim_{x \rightarrow a} f(x)$ exists for all a except $a = -1$. Notice that the right and left limits are different at $a = -1$.

EXAMPLE 4

Question

Sketch the graph of

$$f(x) = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 + x & \text{if } x \geq 1 \end{cases}$$

Then find each of the following or state that it does not exist.

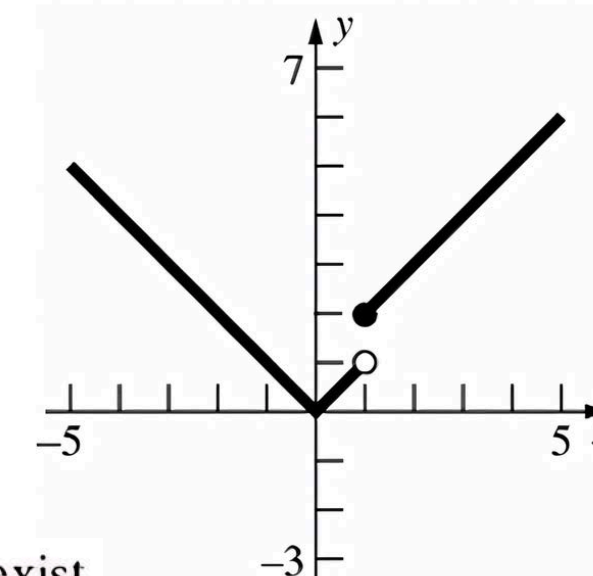
(a) $\lim_{x \rightarrow 0} f(x)$

(b) $\lim_{x \rightarrow 1} f(x)$

(c) $f(1)$

(d) $\lim_{x \rightarrow 1^+} f(x)$

Answered



a. $\lim_{x \rightarrow 0} f(x) = 0$

b. $\lim_{x \rightarrow 1} f(x)$ does not exist

c. $f(1) = 2$

d. $\lim_{x \rightarrow 1^+} f(x) = 2$

Limit

Limit of Trigonometric

Theorem A Limits of Trigonometric Functions

For every real number c in the function's domain,

1. $\lim_{t \rightarrow c} \sin t = \sin c$

3. $\lim_{t \rightarrow c} \tan t = \tan c$

5. $\lim_{t \rightarrow c} \sec t = \sec c$

2. $\lim_{t \rightarrow c} \cos t = \cos c$

4. $\lim_{t \rightarrow c} \cot t = \cot c$

6. $\lim_{t \rightarrow c} \csc t = \csc c$

Theorem B Special Trigonometric Limits

1. $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$

2. $\lim_{t \rightarrow 0} \frac{1 - \cos t}{t} = 0$

EXAMPLE 5

Question

Find each limit.

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x}{x}$$

Answered

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} 3 \frac{\sin 3x}{3x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = \lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$$

Thus,

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 3$$

$$(b) \lim_{t \rightarrow 0} \frac{1 - \cos t}{\sin t}$$

Answered

$$(b) \lim_{t \rightarrow 0} \frac{1 - \cos t}{\sin t} = \lim_{t \rightarrow 0} \frac{\frac{1 - \cos t}{t}}{\frac{\sin t}{t}} = \frac{\lim_{t \rightarrow 0} \frac{1 - \cos t}{t}}{\lim_{t \rightarrow 0} \frac{\sin t}{t}} = \frac{0}{1} = 0$$

$$(c) \lim_{x \rightarrow 0} \frac{\sin 4x}{\tan x}$$

Answered

$$(c) \lim_{x \rightarrow 0} \frac{\sin 4x}{\tan x} = \lim_{x \rightarrow 0} \frac{\frac{4 \sin 4x}{4x}}{\frac{\sin x}{x \cos x}} = \frac{4 \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}}{\left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{1}{\cos x} \right)} = \frac{4}{1 \cdot 1} = 4$$

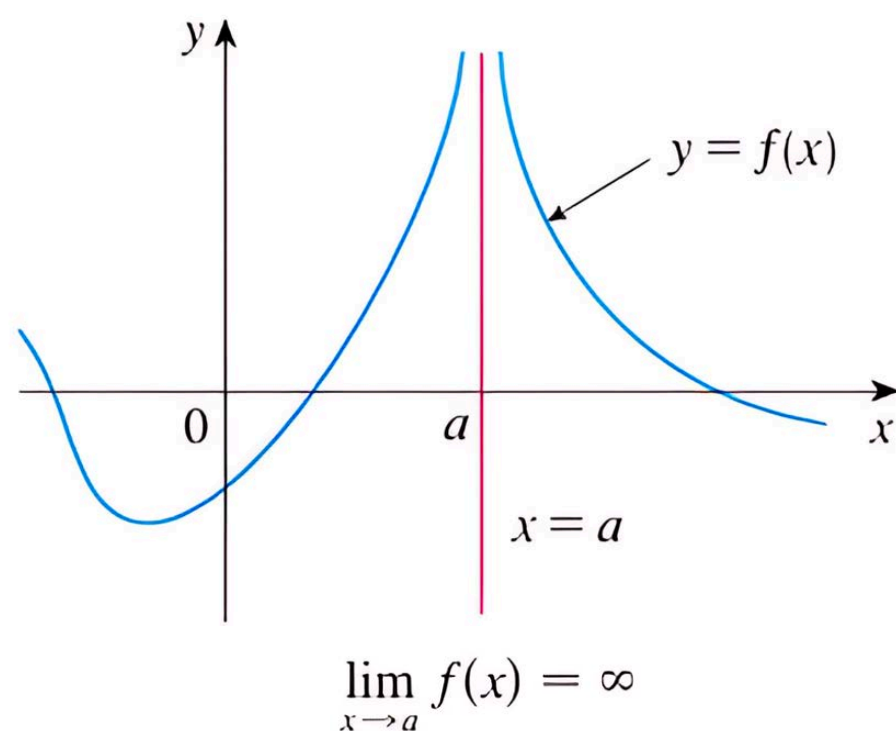
Limit

Infinite Limits

4 Intuitive Definition of an Infinite Limit Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

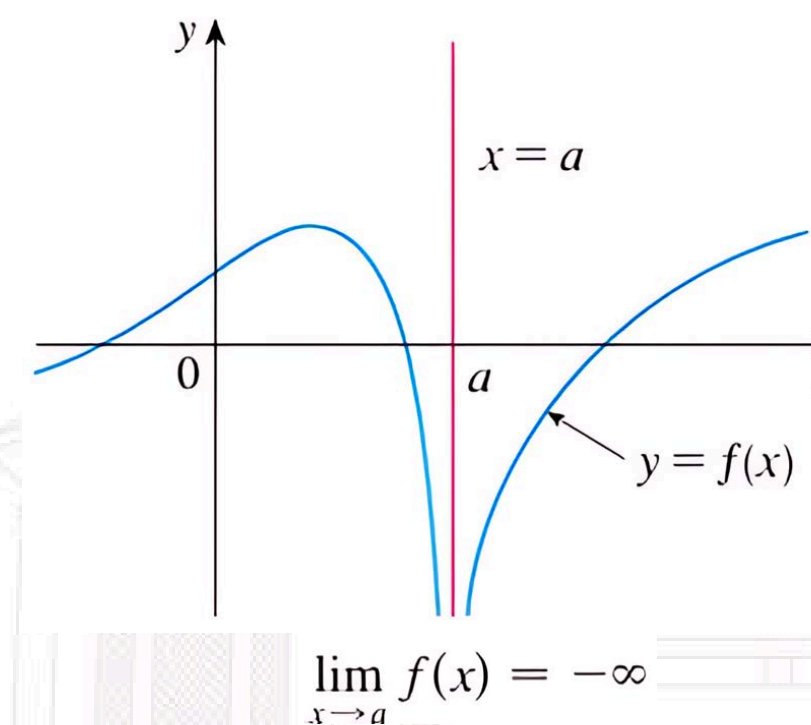
means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .



5 Definition Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a , but not equal to a .



Limit

Infinite Limits

6 Definition The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

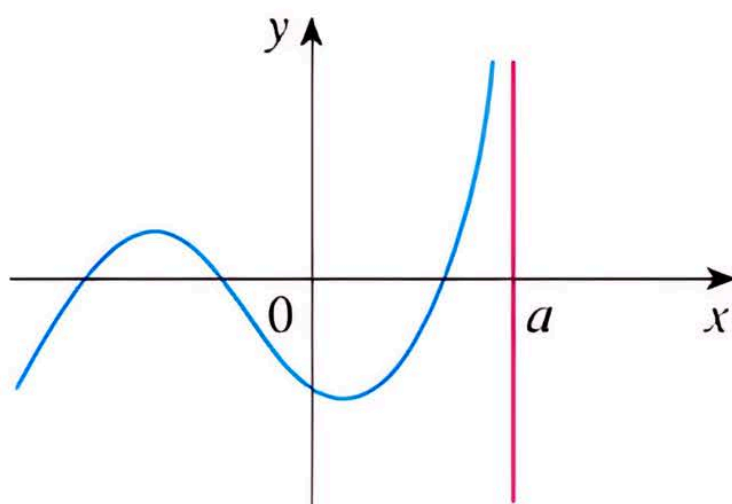
$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

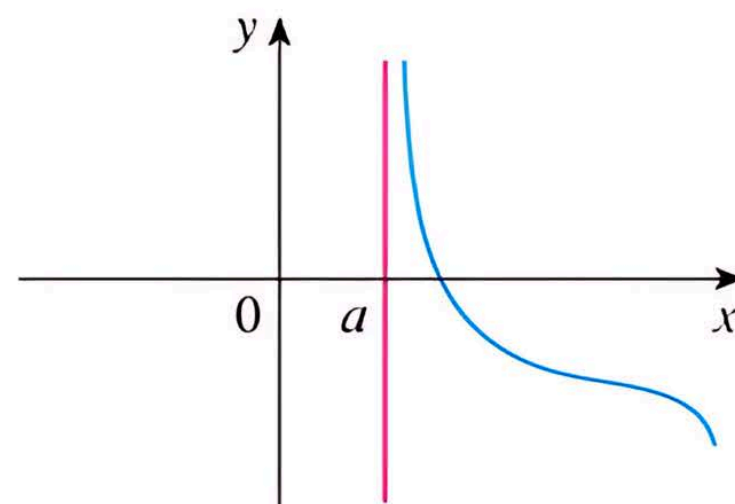
$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

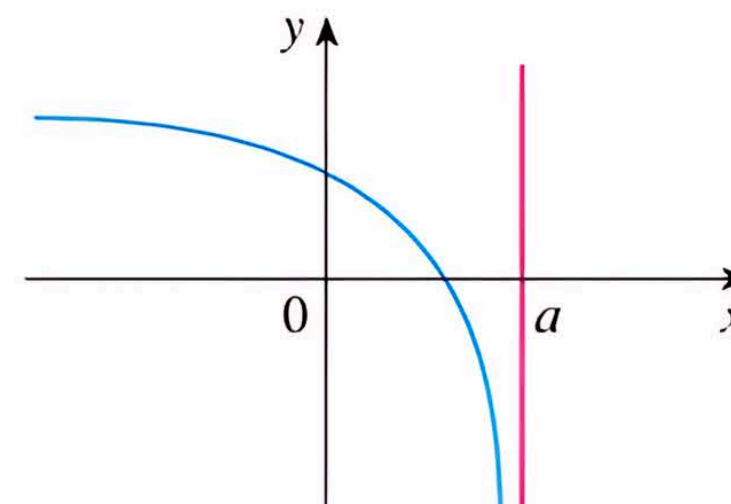
$$\lim_{x \rightarrow \infty} \frac{1}{x^k} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x^k} = 0$$



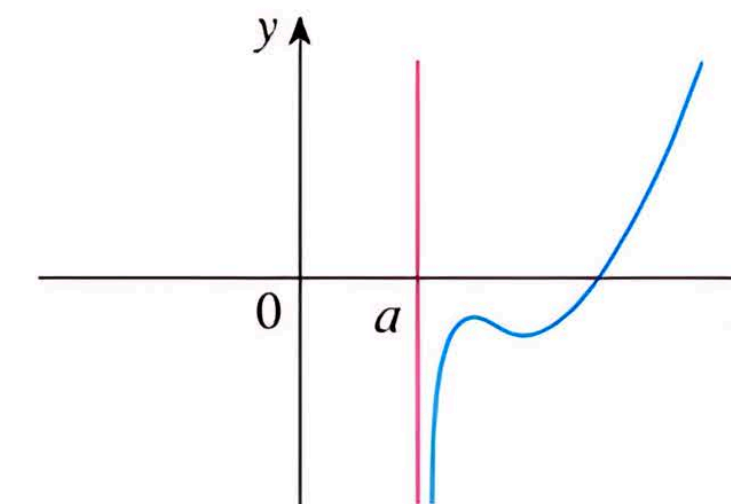
(a) $\lim_{x \rightarrow a^-} f(x) = \infty$



(b) $\lim_{x \rightarrow a^+} f(x) = \infty$



(c) $\lim_{x \rightarrow a^-} f(x) = -\infty$



(d) $\lim_{x \rightarrow a^+} f(x) = -\infty$

Limit

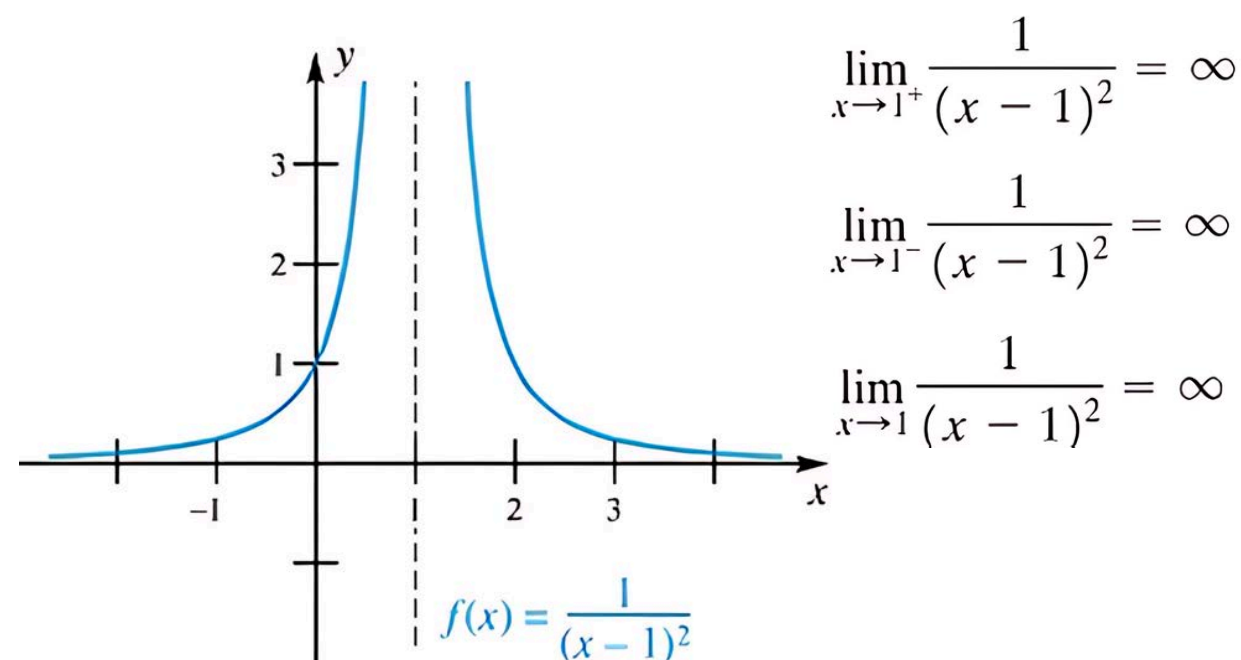
Infinite Limits

EXAMPLE 1

Question

Find $\lim_{x \rightarrow 1^-} \frac{1}{(x-1)^2}$ and $\lim_{x \rightarrow 1^+} \frac{1}{(x-1)^2}$.

Answered



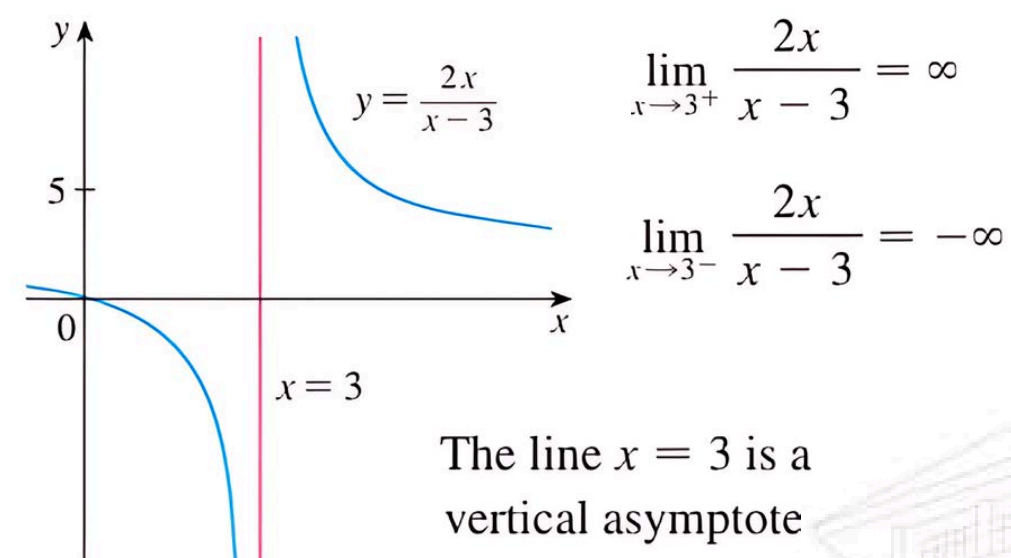
The line $x = 1$
vertical asymptote

EXAMPLE 2

Question

Find $\lim_{x \rightarrow 3^+} \frac{2x}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{2x}{x-3}$

Answered

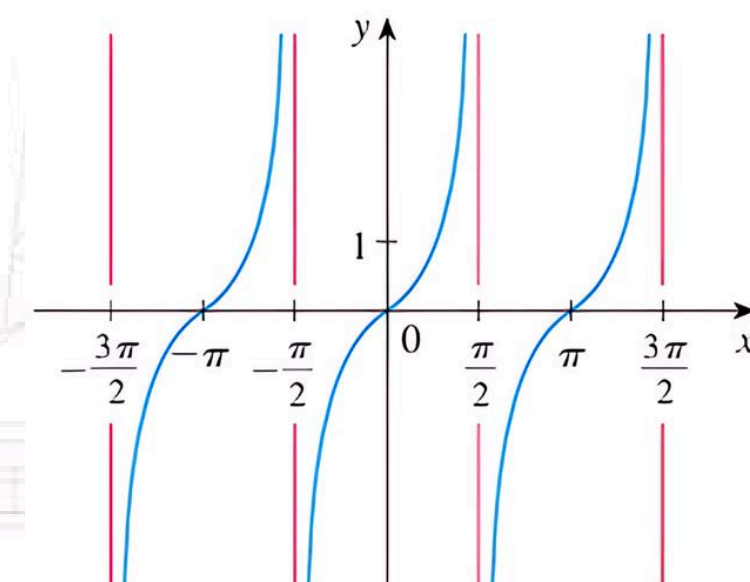


EXAMPLE 3

Question

Find the vertical asymptotes of $f(x) = \tan x$

Answered



$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty \quad \lim_{x \rightarrow (\pi/2)^+} \tan x = -\infty$$

the lines $x = \pi/2 + n\pi$, where n is an integer,
are all vertical asymptotes

EXAMPLE 4

Question

$$\lim_{x \rightarrow \infty} \frac{x^2}{5 - x^3}$$

Answered

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{5}{x^3} - 1} = 0$$

EXAMPLE 5

Question

$$\lim_{t \rightarrow -\infty} \frac{t^2}{7 - t^2}$$

Answered

$$\lim_{t \rightarrow -\infty} \frac{1}{\frac{7}{t^2} - 1} = -1$$

EXAMPLE 6

Question

$$\lim_{t \rightarrow -\infty} \frac{t}{t - 5}$$

Answered

$$\lim_{t \rightarrow -\infty} \frac{1}{1 - \frac{5}{t}} = 1$$

EXAMPLE 7

Question

$$\lim_{\theta \rightarrow -\infty} \frac{\pi \theta^5}{\theta^5 - 5\theta^4}$$

Answered

$$\lim_{\theta \rightarrow -\infty} \frac{\pi}{1 - \frac{5}{\theta}} = \pi$$

Limit

Infinite Limits

EXAMPLE 8

Answered

Question

$$\begin{aligned}\lim_{x \rightarrow \infty} \sqrt{\frac{x^2 + x + 3}{(x-1)(x+1)}} &= \lim_{x \rightarrow \infty} \sqrt{\frac{x^2 + x + 3}{x^2 - 1}} \\ &= \sqrt{\lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x} + \frac{3}{x^2}}{1 - \frac{1}{x^2}}} = \sqrt{1} = 1\end{aligned}$$

EXAMPLE 9

Question

$$\lim_{y \rightarrow -\infty} \frac{9y^3 + 1}{y^2 - 2y + 2}$$

Answered

$$\lim_{y \rightarrow -\infty} \frac{9y^3 + 1}{y^2 - 2y + 2} = \lim_{y \rightarrow -\infty} \frac{9y + \frac{1}{y^2}}{1 - \frac{2}{y} + \frac{2}{y^2}} = -\infty$$

Limit

Limit (Time to participate)

Sketch the graph of

$$g(x) = \begin{cases} -x + 1 & \text{if } x < 1 \\ x - 1 & \text{if } 1 < x < 2 \\ 5 - x^2 & \text{if } x \geq 2 \end{cases}$$

Then find each of the following or state that it does not exist

- (a) $\lim_{x \rightarrow 1} g(x)$ (b) $g(1)$
(c) $\lim_{x \rightarrow 2} g(x)$ (d) $\lim_{x \rightarrow 2^+} g(x)$

1

Sketch the graph of the function and use it to determine the values of a for which $\lim_{x \rightarrow a} f(x)$ exists.

$$f(x) = \begin{cases} 1 + \sin x & \text{if } x < 0 \\ \cos x & \text{if } 0 \leq x \leq \pi \\ \sin x & \text{if } x > \pi \end{cases}$$

2

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^{\sin x}$$

3



SEE YOU NEXT WEEK !

